TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Coefficient of Linear Expansion of a Solid

A brass rod grows by barely two-thirds of a millimetre when steam passes round it — the spherometer's fine screw turns that invisible growth into a number.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 11 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the coefficient of linear expansion α of a metal rod, using a steam jacket to heat it and a spherometer to measure the increase in length. When a solid rod of length L is heated through Δθ, its length increases by ΔL, and experiment shows the increase is proportional both to the original length and to the temperature rise:

$$\Delta L = \alpha\, L\, \Delta\theta \quad \Rightarrow \quad \alpha = \Delta L / (L\, \Delta\theta) \quad \text{per } ^\circ\text{C}$$

α is the fractional increase in length per degree — for brass about 19×10^{-6} per °C, so a half-metre rod heated through some 70 °C grows by roughly 0.65 mm: far too small for a scale, but well within the reach of a spherometer whose least count is 0.001 cm (0.01 mm). The rod stands vertically in the jacket with its lower end on a fixed stop; only the upper end is free to move, and the spherometer's central screw rests on that free end, so the screw's displacement between the cold and hot readings is exactly ΔL. (For reference, the area and volume coefficients of an isotropic solid follow as β ≈ 2α and γ ≈ 3α.)

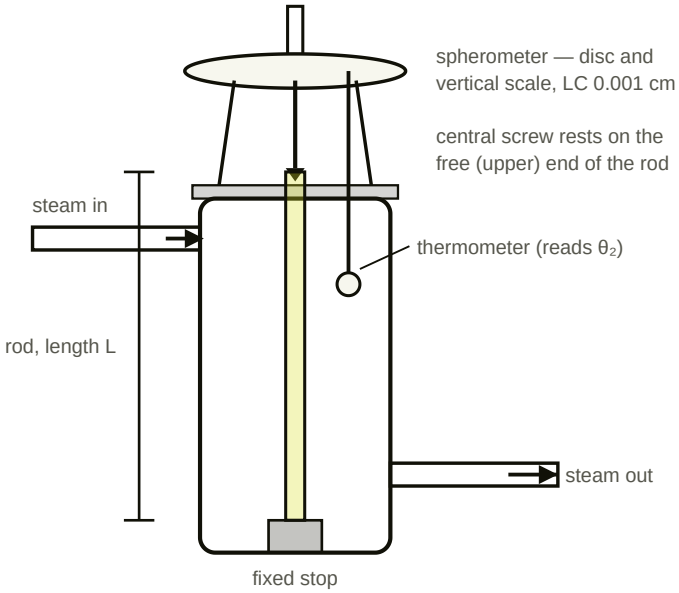


Figure 1 — The steam-jacket apparatus. The rod's lower end sits on a fixed stop inside the jacket; its upper end protrudes through the cork and carries the spherometer's central screw, while the spherometer's legs rest on the fixed cap — so every micrometre the rod grows registers on the disc.

2. Apparatus and procedure

- **Apparatus:** steam jacket with metal rod (brass), spherometer, steam generator (boiler), thermometer, metre scale, rubber tubing.
- Measure the length L of the rod with the metre scale (between the fixed stop and the free end). Note the room temperature θ_1 from the thermometer before any steam is admitted — the rod starts at room temperature throughout.
- Set the spherometer with its legs on the jacket's cap and its central screw just touching the rod's upper end; take the reading (main scale + disc) carefully — this is the cold reading. Note the spherometer's least count (pitch ÷ number of disc divisions).
- Pass steam through the jacket. The rod warms towards the steam temperature; the free end rises and lifts the screw — follow it gently with the screw so contact is kept, and wait until the thermometer in the jacket is steady and the screw reading no longer changes (about 10 minutes). Record the steam temperature θ_2 and the hot spherometer reading.
- The difference of the two readings is ΔL. Compute $\alpha = \Delta L / (L(\theta_2 - \theta_1))$.
- Repeat, if time allows, after the rod has cooled back to room temperature, and average.

3. Worked calculation

Brass rod: L = 50.0 cm. Room temperature θ_1 = 30 °C; steam temperature θ_2 = 98 °C, so $\Delta\theta$ = 68 °C. Spherometer (least count 0.001 cm): cold reading 4.210 mm; hot reading 4.856 mm.

$$\Delta L = 4.856 - 4.210 = 0.646 \text{ mm} = 0.0646 \text{ cm}$$

$$\alpha = 0.0646 / (50.0 \times 68) = 0.0646 / 3400 = 1.90 \times 10^{-5} \text{ per °C}$$

The accepted value for brass is 1.9×10^{-5} per °C — the measurement lands on it, which is the reward for waiting until the reading was truly steady before calling it "hot".

4. Precautions

- Wait for the steady state: a thermometer can read steam temperature while the rod's interior still lags behind it. A reading taken early gives too small a ΔL — and too small an α.
- The spherometer screw must end its motion always in the same rotational sense (approach contact from the same side) to avoid backlash error; "just touching" means the screw tip meets the rod end without pressing it down.
- Do not let condensed water collect in the jacket and touch the rod unevenly; drain the outlet freely. Waterlogging changes the rod's effective heated length.
- Keep hands and breath off the rod and screw while reading — the apparatus measures hundredths of a millimetre, and body heat is not negligible at that scale.

Practice (viva preparation)

1. Why is a spherometer, and not a metre scale or vernier calliper, used to measure ΔL?
2. A steel rail is 12 m long and α for steel is 12×10^{-6} per °C. What expansion gap must be left if the rail temperature rises by 40 °C?
3. The spherometer has a pitch of 1 mm and 100 divisions on its disc. What is its least count, and how many disc divisions correspond to $\Delta L = 0.646$ mm?
4. Why must the spherometer's legs stand on the fixed cap of the jacket rather than on the rod itself?
5. A student records the hot reading while the thermometer is still climbing. Which way does α err, and why?

Answer key

1. The expansion is a fraction of a millimetre (here 0.646 mm). A metre scale resolves 1 mm and a vernier about 0.1 mm — both too coarse. The spherometer's least count of 0.001 cm = 0.01 mm resolves the growth to about 1.5%.
2. $\Delta L = \alpha L \Delta\theta = 12 \times 10^{-6} \times 12 \text{ m} \times 40 = 5.76 \times 10^{-3}$ m ≈ 5.8 mm per rail — the gap that keeps rails from buckling in summer.

3. Least count = pitch/divisions = 1 mm/100 = 0.01 mm = 0.001 cm. The growth 0.646 mm is 64.6 divisions of the disc (plus any whole turns of the screw).
4. The legs define the reference plane. If they stood on the rod, the reference itself would move with the expansion and the screw would record nothing; standing on the fixed cap, only the rod's free end moves relative to them.
5. Too small: the rod has not yet reached the temperature the thermometer is heading for, so the recorded ΔL belongs to a smaller true $\Delta\theta$ than the one used in the formula — wait for both readings to be steady.

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